CSE563 — Unit V Numerical Worksheet

Algorithm Complexity Analysis

40 questions • CO5 • Student Worksheet

Instructions

Attempt all 40 questions. **Never state a complexity without justifying it** — a bare $\Theta(\cdot)$ earns at most half marks. For each recurrence, first decide whether it is *divisive* (n/b, master theorem may apply) or *subtractive* ($n-k$, iterate instead), and say which method you are using. Refer to the *Unit V Numerical Guide* for a worked example of each problem type.

A. Counting Work and Asymptotic Notation (Q1–Q12)

Q1. Give the Θ complexity of the triple loop

```
for i in range(n): for j in range(n): for k in range(n): x += 1
```

Q2. Give the Θ complexity of: `for i in range(n):  for j in range(i+1, n):  x += 1`

Q3. Give the Θ complexity of: `i = n; while i > 0:  i = i // 4`

Q4. Give the Θ complexity of: `for i in range(n):  j = 1; while j < n:  j *= 2`

Q5. Give the Θ complexity of: `for i in range(1, n+1):  for j in range(1, n+1, i):  x += 1`

Q6. Give the Θ complexity of two sequential blocks: one $\Theta(n^2)$ followed by one $\Theta(n \log n)$.

Q7. Prove from the definition that $7n^2 + 5n + 3 = O(n^2)$, giving explicit c and n_0.

Q8. Prove that $7n^2 + 5n + 3 = \Omega(n^2)$, and hence state the tight bound.

Q9. Prove that $2^{n+1} = O(2^n)$ but $2^{2n} \neq O(2^n)$.

Q10. Arrange in increasing order of growth: n^2, $n \log n$, 2^n, $\log n$, $n!$, n, $\sqrt{n}$, $n^{1.5}$, 1.

Q11. Use the limit test to determine whether $f(n) = n \log n$ is O, Θ or Ω of $g(n) = n^2$.

Q12. An algorithm performs $T(n) = 4n^3 + 1000n^2 + 10^6$ operations. State the Θ class, and explain why the huge constant 10^6 is irrelevant asymptotically but may matter in practice.

B. Recurrence Relations (Q13–Q24)

Q13. Write the recurrence for binary search and for merge sort, and state the solution of each.

Q14. Solve $T(n) = T(n-1) + c$ with $T(1) = c$, by iteration.

Q15. Solve $T(n) = T(n-1) + n^2$ with $T(0) = 0$, by iteration.

Q16. Solve $T(n) = 2T(n-1) + 1$ with $T(1) = 1$ (Tower of Hanoi), by iteration.

Q17. Solve $T(n) = 2T(n-1) + 3$ with $T(0) = 1$, by iteration, and verify at $n = 3$.

Q18. Use a recursion tree to solve $T(n) = 3T(n/3) + n$.

Q19. Use a recursion tree to solve $T(n) = 2T(n/2) + n^2$, and explain which level dominates.

Q20. Prove by substitution (induction) that $T(n) = T(n/2) + 1$ is $O(\log n)$.

Q21. Solve the linear recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_0 = 1$, $a_1 = 0$, and verify a_2.

Q22. Solve $a_n = 4a_{n-1} - 4a_{n-2}$ with $a_0 = 1$, $a_1 = 6$ (note the repeated root).

Q23. State the Fibonacci recurrence, solve its characteristic equation, and explain why naive recursive Fibonacci is exponential.

Q24. For $T(n) = T(n-1) + \log n$, find the Θ class.

C. The Master Theorem (Q25–Q30)

Q25. Apply the master theorem to $T(n) = 9T(n/3) + n^2$, naming the case.

Q26. Apply the master theorem to $T(n) = 16T(n/4) + n$, naming the case.

Q27. Apply the master theorem to $T(n) = 3T(n/4) + n \log n$, including the regularity check.

Q28. Apply the master theorem to Strassen's recurrence $T(n) = 7T(n/2) + n^2$ and to Karatsuba's $T(n) = 3T(n/2) + n$, comparing each with the naive algorithm.

Q29. Explain why the master theorem cannot be applied to $T(n) = 2T(n/2) + n \log n$, and give the correct answer.

Q30. State two further recurrences to which the master theorem does not apply, giving the reason in each case.

D. Complexity Classes and SAT (Q31–Q35)

Q31. Classify each as P or NP-complete, with a one-line justification: (a) sorting, (b) Hamiltonian cycle, (c) Euler circuit, (d) 3-colouring, (e) shortest path with non-negative weights.

Q32. Define P, NP, NP-hard and NP-complete, and state the relationship between them.

Q33. State precisely the two steps needed to prove a problem X is NP-complete, and explain why the reduction must go in a particular direction.

Q34. Determine whether $\Phi = (a \vee b \vee c) \wedge (\neg a \vee \neg b) \wedge (\neg b \vee \neg c) \wedge (\neg a \vee \neg c)$ is satisfiable. If so, give a satisfying assignment.

Q35. Show that $\Psi = (x \vee y) \wedge (\neg x \vee y) \wedge (x \vee \neg y) \wedge (\neg x \vee \neg y)$ is unsatisfiable, using resolution.

E. Searching and Sorting (Q36–Q40)

Q36. Trace binary search for the key 56 in $A = [3, 9, 14, 22, 31, 45, 56, 68, 79, 88, 95]$ (indices 0–10). State the number of comparisons and the worst case for $n = 11$.

Q37. Trace the Lomuto partition of $A = [8, 3, 5, 10, 2, 7]$ with the last element as pivot. Give the array after each swap and the final pivot index.

Q38. Show the merge sort of $A = [38, 27, 43, 3, 9, 82, 10]$, giving the splitting and the merging steps.

Q39. State the best, average and worst case of quicksort, describe the input that triggers the worst case with a last-element pivot, and state how it is avoided.

Q40. Prove that comparison-based sorting requires $\Omega(n \log n)$ comparisons in the worst case, and name one algorithm that sorts in linear time, stating the assumption that permits it.